

Theory companion: how the pieces fit together

Coupled magnetic–mechanical oscillator — derivations, dependencies, and what goes where

0. First: why the beat envelope in the earlier figure was “shifted”

It was a sign error in the *plotting script*, not in the physics; the figure is now regenerated (Figure 2 below). The lesson is worth keeping because it is exactly the calculation you need for the beat-minimum test. With $x_1 = \frac{1}{2}(\eta_0 \cos \omega_1 t - \xi_0 \cos \omega_2 t)$, the amplitude of the fast oscillation is

$$|x_1|_{\text{env}} = \frac{1}{2} \sqrt{\eta_0^2 + \xi_0^2 - 2\eta_0 \xi_0 \cos(\omega_2 - \omega_1)t},$$

and for a hold-and-release start $\eta_0 = -(A + x_{20}) < 0$ while $\xi_0 = A - x_{20} > 0$. So at $t = 0$ the square root is $\sqrt{(|\eta_0| + \xi_0)^2}$, i.e. the envelope starts at its *maximum* A (the spring was just released from $-A$) and reaches its first *minimum* x_{20} at $t = T_{\text{beat}}/2$. The old script had $+2\eta_0 \xi_0$, which put the minimum at $t = 0$ — a quarter-beat shift. Rule: derive the envelope from the actual signs of η_0 and ξ_0 , never from “cos + cos” memory.

1. The chain of reasoning (main text)

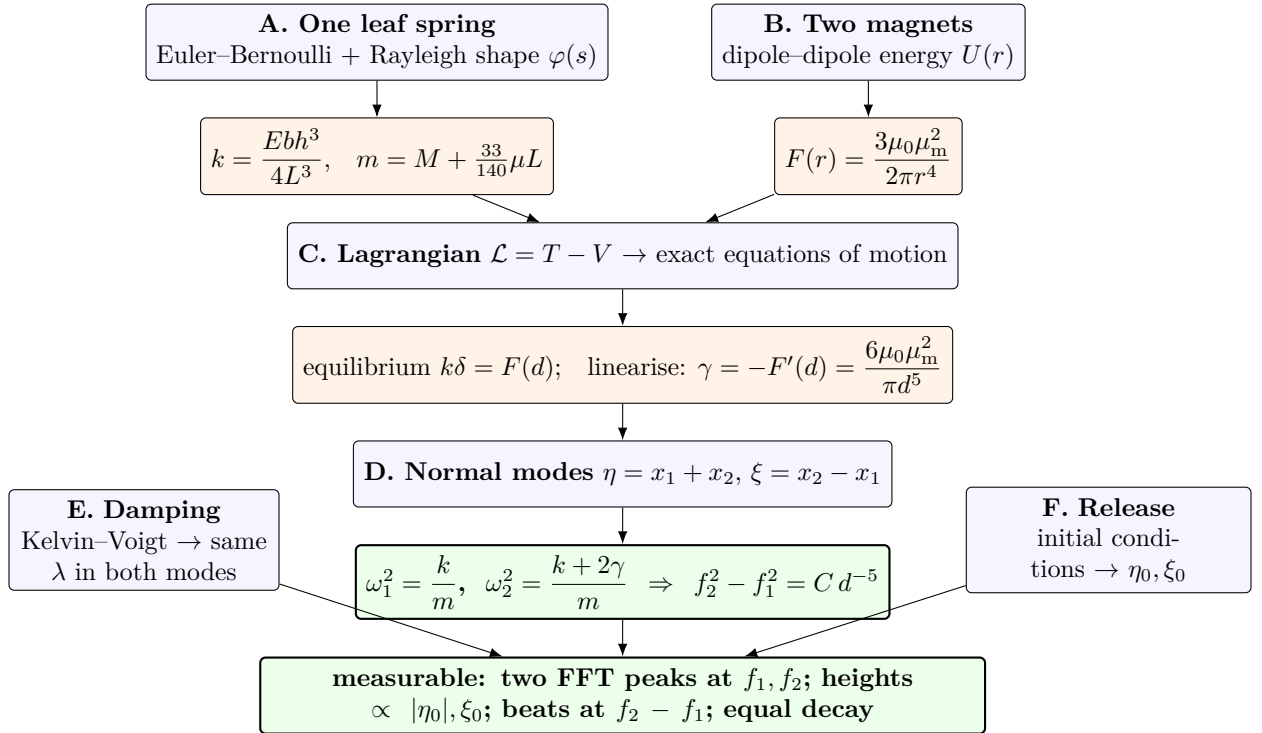


Figure 1: Every box feeds the next; nothing is assumed that is not derived above it. The extension (Section 2) branches off at C: keep $F(r)$ exact instead of linearising.

A. One leaf spring $\rightarrow k$ and m

A1. Shape function (the one modelling assumption)

The strip vibrates in its lowest mode, whose shape is taken as the static deflection curve under a tip load:

$$y(s, t) = x(t) \varphi(s), \quad \varphi(s) = \frac{3Ls^2 - s^3}{2L^3}.$$

Check the boundary conditions: $\varphi(0) = 0$ (clamped), $\varphi'(0) = 0$ (clamped, no slope), $\varphi(L) = 1$ (tip moves by x), $\varphi''(L) = 0$ (no bending moment at a free end). One function of s times one function of t : the beam becomes a single coordinate $x(t)$.

A2. Energies $\rightarrow m$ and k

Kinetic energy of the strip: each element μds moves at $\dot{x}\varphi(s)$, so

$$T_{\text{beam}} = \frac{1}{2} \mu \dot{x}^2 \int_0^L \varphi^2 ds = \frac{1}{2} \left(\frac{33}{140} \mu L \right) \dot{x}^2.$$

Elastic energy of a bent beam is $\frac{1}{2} EI \int (y'')^2 ds$ (curvature y'' , flexural rigidity EI , $I = bh^3/12$):

$$V_{\text{beam}} = \frac{1}{2} EI x^2 \int_0^L (\varphi'')^2 ds = \frac{1}{2} \left(\frac{3EI}{L^3} \right) x^2.$$

The two integrals are $\int_0^L \varphi^2 ds = \frac{33}{140} L$ and $\int_0^L \varphi''^2 ds = 3/L^3$ (with $\varphi'' = 3(L-s)/L^3$). Adding the magnet ($\frac{1}{2} M \dot{x}^2$):

$$m = M + \frac{33}{140} \mu L, \quad k = \frac{3EI}{L^3} = \frac{Ebh^3}{4L^3}$$

Used in: C (as the T and V of each spring). *By-products:* tip rotation $\theta = \varphi'(L)x = 3x/2L$ (used in B for the tilt correction); the $\frac{33}{140}$ is why the magnet mass alone is not m .

Pitfalls

$T_{\text{beam}} = \frac{33}{280} \mu L \dot{x}^2$ and $m_{\text{eff}} = \frac{33}{140} \mu L$ are the same statement ($T = \frac{1}{2} m_{\text{eff}} \dot{x}^2$). Quote one, not both as if different. $k = 3EI/L^3$ is the tip stiffness of a cantilever — don't confuse with $48EI/L^3$ (simply supported beam).

B. Two magnets $\rightarrow F(r)$

B1. Dipole energy

Magnet 2 (moment $\boldsymbol{\mu}_2$) in the field $\mathbf{B}_1$ of magnet 1: $U = -\boldsymbol{\mu}_2 \cdot \mathbf{B}_1$ with $\mathbf{B}_1 = \frac{\mu_0}{4\pi r^3} [3(\boldsymbol{\mu}_1 \hat{\mathbf{r}}) \hat{\mathbf{r}} - \boldsymbol{\mu}_1]$, hence

$$U = \frac{\mu_0}{4\pi r^3} [\boldsymbol{\mu}_1 \cdot \boldsymbol{\mu}_2 - 3(\boldsymbol{\mu}_1 \cdot \hat{\mathbf{r}})(\boldsymbol{\mu}_2 \cdot \hat{\mathbf{r}})].$$

Like poles facing on a common axis: $\boldsymbol{\mu}_1 = \mu_m \hat{\mathbf{r}}$, $\boldsymbol{\mu}_2 = -\mu_m \hat{\mathbf{r}}$, so $\boldsymbol{\mu}_1 \cdot \boldsymbol{\mu}_2 = -\mu_m^2$ and $(\boldsymbol{\mu}_1 \cdot \hat{\mathbf{r}})(\boldsymbol{\mu}_2 \cdot \hat{\mathbf{r}}) = -\mu_m^2$:

$$U(r) = \frac{\mu_0}{4\pi r^3} [-\mu_m^2 + 3\mu_m^2] = \frac{\mu_0 \mu_m^2}{2\pi r^3} > 0.$$

B2. Force and its slope

$$F(r) = -\frac{dU}{dr} = \frac{3\mu_0\mu_m^2}{2\pi r^4} \text{ (repulsive)}, \quad \gamma(r) \equiv -\frac{dF}{dr} = \frac{6\mu_0\mu_m^2}{\pi r^5} = \frac{4F(r)}{r}.$$

Used in: C (equilibrium uses $F(d)$; linearisation uses γ). *Tilt (optional, one sentence)*: with the tips rotated by $\theta_i = 3x_i/2L$ the bracket becomes $2\cos\theta_1\cos\theta_2 - \sin\theta_1\sin\theta_2$; expanding to second order changes both mode frequencies by relative amounts $\lesssim \frac{9}{32}\frac{\gamma}{k}(d/L)^2 < 10^{-3}$. Say it is negligible and move on.

C. Lagrangian $\rightarrow$ equations of motion, equilibrium, linearisation

C1. Lagrangian and exact equations

Coordinates: x_1, x_2 = tip displacements from the springs' *unstressed* positions, both positive to the right; separation $r = d_0 + x_2 - x_1$.

$$\mathcal{L} = \frac{1}{2}m(\dot{x}_1^2 + \dot{x}_2^2) - \frac{1}{2}k(x_1^2 + x_2^2) - U(d_0 + x_2 - x_1).$$

$$\frac{d}{dt}\frac{\partial\mathcal{L}}{\partial\dot{x}_i} = \frac{\partial\mathcal{L}}{\partial x_i} \text{ with } \partial U/\partial x_1 = -U'(r) = +F, \partial U/\partial x_2 = U'(r) = -F:$$

$$m\ddot{x}_1 = -kx_1 - F(r), \quad m\ddot{x}_2 = -kx_2 + F(r).$$

(Read: the repulsion pushes 1 to the left and 2 to the right. Exact within the dipole and one-mode assumptions.)

C2. Equilibrium

Set accelerations to zero: $kx_1 = -F$, $kx_2 = +F \Rightarrow$ each spring bent outwards by $\delta = F(d)/k$, separation $d = d_0 + 2\delta$. *This d (with magnets mounted) is the measured variable.*

C3. Linearisation

Now let x_1, x_2 be displacements from equilibrium, so $r = d + \xi$ with $\xi = x_2 - x_1$. Taylor: $F(d + \xi) \simeq F(d) + F'(d)\xi = F(d) - \gamma\xi$. Insert; the constant $F(d)$ cancels against $k\delta$:

$$m\ddot{x}_1 = -kx_1 + \gamma(x_2 - x_1), \quad m\ddot{x}_2 = -kx_2 - \gamma(x_2 - x_1)$$

Interpretation: a compressed spring of stiffness γ between the magnets. *Validity*: $|\xi| \ll d$; the neglected term is $\frac{1}{2}F''(d)\xi^2 = \frac{5\gamma}{2d}\xi^2$ — this is the entry point of the extension.

D. Normal modes $\rightarrow$ the tested relation

D1. Decoupling

Add and subtract the two equations:

$$m\ddot{\eta} = -k\eta, \quad m\ddot{\xi} = -(k + 2\gamma)\xi, \quad \eta \equiv x_1 + x_2, \quad \xi \equiv x_2 - x_1.$$

Two independent harmonic oscillators:

$$\omega_1^2 = \frac{k}{m} \quad (\text{in-phase: } \xi = 0), \quad \omega_2^2 = \frac{k + 2\gamma}{m} \quad (\text{anti-phase: } \eta = 0)$$

Why 2γ : in the anti-phase mode each magnet moves a , the gap changes by $2a$, the extra force on each is $\gamma \cdot 2a$.

D2. Eliminate the unknown spring

$$\omega_2^2 - \omega_1^2 = \frac{2\gamma}{m} \Rightarrow f_2^2 - f_1^2 = \frac{\gamma}{2\pi^2 m} = \frac{6\mu_0\mu_m^2}{2\pi^3 m} d^{-5} = \underbrace{\frac{3\mu_0\mu_m^2}{\pi^3 m}}_C d^{-5}.$$

$$\boxed{\ln(f_2^2 - f_1^2) = \ln C - 5 \ln d}$$

Predictions to test: (i) f_1 independent of d and equal to the single-spring frequency; (ii) gradient -5 ; (iii) intercept gives C , i.e. μ_m if m is known. Everything unknown about the spring has dropped out of the gradient.

E. Damping (Kelvin–Voigt) → why the frequencies are safe and what the envelope tests

E1. From the material law to the mode equations

Kelvin–Voigt solid: $\sigma = E(\varepsilon + \tau_v \dot{\varepsilon})$. Integrated over the cross-section, the bending moment becomes $EI(y'' + \tau_v \dot{y}'')$, so the beam equation is

$$\mu \ddot{y} = -EI(y'''' + \tau_v \dot{y}'''').$$

Insert $y = x(t)\varphi(s)$ and project onto φ (multiply by φ , integrate — the same two integrals as in A2): $m\ddot{x} = -kx - k\tau_v \dot{x}$. Air drag on the magnet adds $-c_a \dot{x}$. Both are linear in $\dot{x}$ and identical for identical springs, so in the mode equations

$$\ddot{\eta} + 2\lambda\dot{\eta} + \omega_1^2\eta = 0, \quad \ddot{\xi} + 2\lambda\dot{\xi} + \omega_2^2\xi = 0, \quad 2\lambda = \frac{k\tau_v + c_a}{m}.$$

E2. Consequences

Solutions $e^{-\lambda t} \cos(\omega'_i t + \phi)$ with $\omega'_i = \sqrt{\omega_i^2 - \lambda^2}$. With $\tau_d = 1/\lambda \approx 21$ s and $\omega \approx 41$ s⁻¹: $\lambda/\omega \approx 10^{-3}$, frequency shift $\lambda^2/2\omega^2 \approx 10^{-6}$ — negligible. Peak full width $\approx 1/\pi\tau_d \approx 15$ mHz. Both modes decay at the *same* rate $\Rightarrow$ ratio of the two FFT peaks and ratio of beat minima to maxima are constant in time. A relative-motion damping (eddy currents between the magnets) would break this — a testable statement.

F. The release → what the record should look like

F1. Initial conditions

Spring 1 held at $x_1 = -A$, spring 2 free and at rest: from the static form of C3, $kx_2 = \gamma(x_1 - x_2) \Rightarrow x_2 = -\frac{\gamma}{k + \gamma}A \equiv -x_{20}$. Both released from rest, so only cosines appear:

$$\eta_0 = x_1(0) + x_2(0) = -(A + x_{20}), \quad \xi_0 = x_2(0) - x_1(0) = A - x_{20}.$$

F2. Motion, spectrum, beats

$$x_1 = \frac{1}{2}(\eta_0 \cos \omega_1 t - \xi_0 \cos \omega_2 t), \quad x_2 = \frac{1}{2}(\eta_0 \cos \omega_1 t + \xi_0 \cos \omega_2 t).$$

Spectrum: two lines at f_1, f_2 in *both* records, heights $\frac{1}{2}|\eta_0|$ and $\frac{1}{2}\xi_0$, ratio

$$\frac{\xi_0}{|\eta_0|} = \frac{A - x_{20}}{A + x_{20}} = \frac{k}{k + 2\gamma} = \left(\frac{f_1}{f_2}\right)^2.$$

(Algebra: $A - x_{20} = A \frac{k}{k+\gamma}$, $A + x_{20} = A \frac{k+2\gamma}{k+\gamma}$.) *Beats:* write $x_1 = \Re[\frac{1}{2}(\eta_0 e^{i\omega_1 t} - \xi_0 e^{i\omega_2 t})]$; the modulus of the bracket is the envelope,

$$|x_1|_{\text{env}} = \frac{1}{2} \sqrt{\eta_0^2 + \xi_0^2 - 2\eta_0 \xi_0 \cos(\omega_2 - \omega_1)t} \in [x_{20}, A], \quad T_{\text{beat}} = \frac{1}{f_2 - f_1}.$$

Maximum A at $t = 0$ (just released), minimum x_{20} at $t = T_{\text{beat}}/2$; spring 2 is the mirror image (its envelope has $+2\eta_0 \xi_0$): energy exchange. Three independent handles on γ/k : peak ratio, beat minimum, and f_2/f_1 itself.

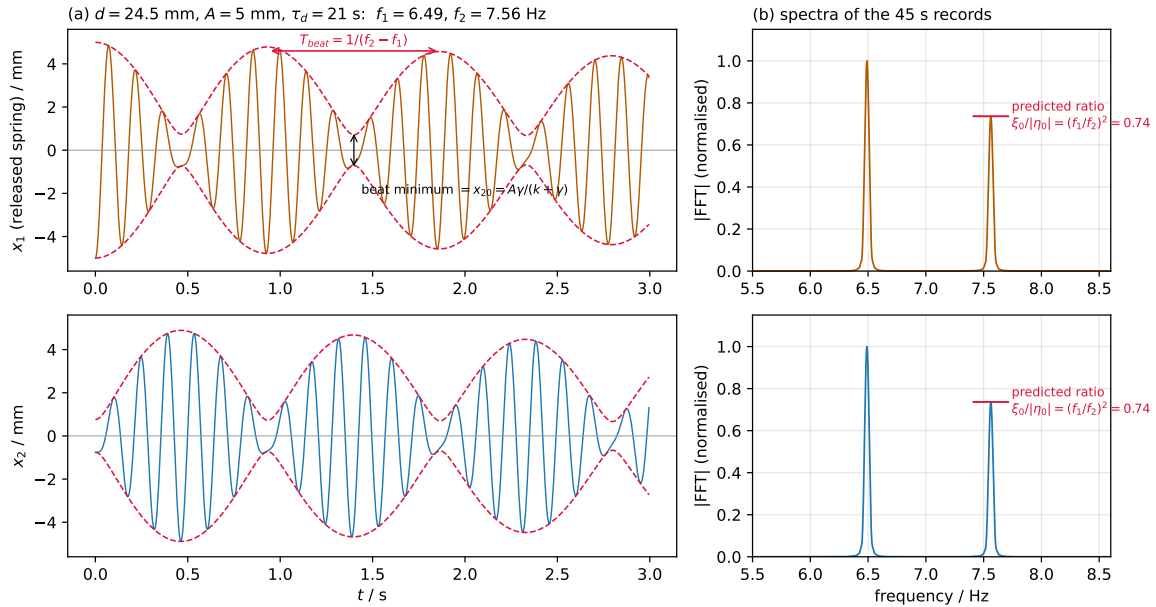


Figure 2: Corrected figure: envelope from F2 with the actual signs of η_0, ξ_0 ; it starts at A and first touches x_{20} at $T_{\text{beat}}/2$; the second spring is a quarter-beat behind.

What to write in the essay

Main-text theory, in this order and roughly this length (about 3–4 pages with figures):

1. **One spring** (i page + Fig. “geometry of the leaf spring”): A1–A2 boxed results; one sentence on why Rayleigh’s shape is adequate ($< 0.2\%$ error with a tip mass).
2. **Two magnets** (i page + Fig. “equilibrium and coordinates”): B1–B2; state $F \propto r^{-4}$ and $\gamma = 4F/d$.
3. **Lagrangian and small oscillations** (i page): C1–C3; the boxed linear equations; state the validity condition $|\xi| \ll d$ and that the extension removes it.
4. **Normal modes** (i page + Fig. “two modes”): D1–D2 with the key box $f_2^2 - f_1^2 = Cd^{-5}$ and the three predictions.
5. **Damping and release** (i page + Fig. 2): E2 and F2 as statements with the two-line derivations; the summary table of predictions vs. tests.

Every box’s “Used in” line is the sentence that links the paragraphs — write those transitions explicitly; that is what makes the theory read as an argument rather than a list.

2. Extension: the non-linear force law

X1. Exact mode equations (branch off at C1, do not linearise)

Subtract and add the exact equations of C1, with x_i measured from equilibrium ($r = d + \xi$):

$$m\ddot{\eta} = -k\eta \quad (\text{exactly}), \quad m\ddot{\xi} = -k\xi + 2[F(d + \xi) - F(d)].$$

Sign check: $\xi > 0$ (further apart) $\Rightarrow F(d + \xi) < F(d) \Rightarrow$ bracket negative $\Rightarrow$ restoring. Linear limit: $2F'(d)\xi = -2\gamma\xi$. **Key statement for the essay:** the centre of mass is exactly harmonic at ω_1 whatever the force law, because the magnetic force is internal; all non-linearity is in the relative coordinate — so f_1 can never depend on amplitude, and only f_2 can.

X2. Expand

$F(d + \xi) = F(d)(1 + u)^{-4}$, $u = \xi/d$, and $(1 + u)^{-4} = 1 - 4u + 10u^2 - 20u^3 + \dots$; with $2F(d)/d = \gamma/2$:

$$\ddot{\xi} + \omega_2^2 \xi = \frac{\gamma}{m} \left[5 \frac{\xi^2}{d} - 10 \frac{\xi^3}{d^2} + \dots \right] = \omega_2^2 g \left[5 \frac{\xi^2}{d} - 10 \frac{\xi^3}{d^2} \right], \quad g \equiv \frac{\gamma}{k + 2\gamma} = \frac{\gamma/m}{\omega_2^2}.$$

Two small parameters: g (how magnetic the anti-phase stiffness is; 0.13 at 24.5 mm) and ξ/d (amplitude).

X3. Solve by successive approximation (Lindstedt–Poincaré)

Try $\xi = a \cos \omega t +$ (small corrections) and use $\cos^2 = \frac{1}{2}(1 + \cos 2\omega t)$, $\cos^3 = \frac{1}{4}(3 \cos \omega t + \cos 3\omega t)$.

Second order (ξ^2 term drives 0 and 2ω): $\xi_2 = \frac{5ga^2}{2d} - \frac{5ga^2}{6d} \cos 2\omega t$ (the $\cos 2\omega t$ response is divided by $\omega_2^2 - 4\omega^2 \simeq -3\omega_2^2$, hence the $1/6$ and the minus sign). *Third order*: the $\cos \omega t$ terms produced by ξ^3 and by $\xi_1 \xi_2$ would grow without bound (resonant) unless the frequency shifts; demanding they cancel gives

$$\frac{\omega - \omega_2}{\omega_2} = \left(\frac{a}{d} \right)^2 \left[\frac{15g}{4} - \frac{125g^2}{12} \right], \quad \xi_3 \ni \left(\frac{5g}{16} + \frac{25g^2}{48} \right) \frac{a^3}{d^2} \cos 3\omega t.$$

(All four coefficients checked by computer algebra and by direct numerical integration — agreement to 2 % at $a/d = 0.17$.)

X4. What it predicts, numerically ($d = 24.5$ mm, $A = 5$ mm, pilot constants)

$g = 0.13$, $a \simeq \xi_0 = 4.2$ mm: f_2 up by 0.9 % (0.07 Hz — comparable to a frequency bin of a 10 s record, so real but small); second harmonic $2f_2$ at 1.9 % of the fundamental in the spectrum of ξ ; mean separation increased by 0.24 mm; f_1 unchanged. With $A = 10$ mm multiply the shift by 4. In a decaying record the FFT sees a time-average, roughly 0.15–0.2 of the initial-amplitude shift.

X5. Internal resonance (the “critical value”)

If $\omega_2 = 2\omega_1$, the $2\omega_2$ -driven terms are far from resonance but the η – ξ cross terms (from tilt or any small asymmetry) become resonant and energy moves between modes: the 2:1 internal resonance other groups mention. Condition: $1 + 2\gamma/k = 4 \Rightarrow 2\gamma/k = 3 \Rightarrow d^* = (C/3f_1^2)^{1/5} \approx 16$ mm — below the smallest achievable d and outside the dipole regime. State that it is not met and that this is why the linear model stays valid throughout the range.

What to write in the essay

Extension / evaluation (1–1½ pages): X1 (three lines — the exact- η statement is the strongest single sentence in the extension), X2 (the expanded equation), X3 boxed results only, X4 as a table of numbers at your two smallest d , X5 as one paragraph. Then use it: (i) explain any upward bend of the $\ln\text{--}\ln$ residuals at small d ; (ii) justify the amplitude control; (iii) if you have an amplitude series at fixed d , plot f_2 against ξ_0^2 — the gradient tests X3 directly; (iv) look for the $2f_2$ line in the spectrum of ξ (log scale) as an optional check.

3. Checks performed (so you can quote them)

quantity	result	how checked
$\int_0^L \varphi^2 ds, \int_0^L \varphi''^2 ds, \varphi'(L)$	$\frac{33}{140}L, 3/L^3, 3/2L$	symbolic integration
Rayleigh ω_0 vs exact cantilever+tip mass	+1.5 % ($M = 0$), < 0.2 % ($M > \frac{1}{2}\mu L$)	transcendental root of exact solution
$F = 3\mu_0\mu_m^2/2\pi r^4, \gamma = 6\mu_0\mu_m^2/\pi r^5 = 4F/r$		symbolic differentiation
tilt correction to ω_1^2, ω_2^2	$-\frac{9}{32}\frac{\gamma}{k}(d/L)^2, -\frac{3}{32}g(d/L)^2$	2nd-order series of the tilted-dipole energy
$(1+u)^{-4}$ coefficients	1, -4, 10, -20	series
Lindstedt coefficients ($c_0, c_2, c_3, \Delta\omega$)	as in X3	symbolic projection onto 1, $\cos 2\tau, \cos \tau, \cos 3\tau$
X3 vs numerical integration	shift, c_2, c_3, c_0 within 2–5 % up to $a/d = 0.17$	nonlinear_check.py
envelope of x_1	starts at A , minimum x_{20} at $T_{\text{beat}}/2$	Hilbert transform of the synthetic signal
peak ratio $\xi_0/ \eta_0 = (f_1/f_2)^2$	0.737 = 0.737 at 24.5 mm	synthetic spectrum

Pitfalls

- Sign conventions: both x_i positive to the right; $\xi = x_2 - x_1$ is the *change of separation*; $\eta_0 < 0$ for a leftward pull. Keep one convention for the whole essay.
- $f_{\text{beat}} = f_2 - f_1$ (period of the envelope swellings), not $(f_2 - f_1)/2$ (period of the amplitude *sign* pattern). Your RQ uses the former.
- Kelvin–Voigt is a statement about the *strip*; it predicts equal λ for both modes only because the strips are identical and the coupling is conservative. Say that explicitly.
- C contains m (the effective mass), not M ; and μ_m is the dipole moment of one magnet ($= B_r V/\mu_0$ for a uniformly magnetised block, if you want a data-sheet estimate).
- “The dipole approximation” is an assumption of B, not of C or D; if the exponent comes out below 5, the suspect is B (finite magnet size), not the normal-mode algebra.